

SE LAB-3 REPORT

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Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Reason 1 — The system's concerns map cleanly onto layers. The kiosk has three distinct responsibilities: touch-screen interaction, order and payment logic, and hardware/data access. A layered structure isolates each one, so the receipt printer driver can be replaced, or menu pricing moved to a remote source, without touching `OrderManager`'s logic.

Reason 2 — The deployment scale does not justify a distributed style. The kiosk serves one customer at a time on one machine. Microservices would add network latency and operational overhead with no scaling benefit. Client-server would make a remote server a single point of failure. Layered keeps the whole flow local and deployable as one unit.

Security Advantage

Card data is confined to `PaymentService` in the business layer. `KioskUI` never touches card details — `OrderManager` receives only an authorisation result via `IPaymentProcessing`, narrowing the attack surface exposed by the public touch interface.

Performance Benefit

All inter-component calls are in-process, avoiding network round trips. `MenuDatabase` can cache menu/pricing at startup, making menu rendering instant — important for queue speed in a busy café.